AI Agents and Agentic AI with Python & Generative AI

Module 1
Agentic AI Concepts

- Introduction
 Video • 10 min
- Flipped Interaction Pattern
 Video • 7 min
- The Agent Loop
 Video • 8 min
- Knowledge Check: Agent Loop & Flipped Interaction
 Practice Assignment • Grade: 100%
- Running the Code Samples in the Course
 Reading • 10 min
- Programmatic Prompting for Agents
 Ungraded Plugin • 15 min
- Try Out Programmatic Prompting
 Reading • 10 min
- Programmatic Prompting for Agents II
 Ungraded Plugin • 15 min
- Programmatic Prompting for Agents III
 Ungraded Plugin • 15 min
- **Programmatic Prompting: Building Blocks for AI Agents**
 Dialogue • 15 min
- Try Out the Customer Service Agent
 Reading • 10 min
- Giving Agents Memory
 Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents
 Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents (Solution)
 Ungraded Plugin • 15 min
- Adding Structure to AI Agent Outputs
 Video • 14 min
- Building Your First Agent
 Ungraded Plugin • 15 min
- AI Agent / Environment Interface
 Ungraded Plugin • 15 min
- AI Agent Feedback and Memory
 Ungraded Plugin • 15 min
- The Agent Loop: From Understanding to Thinking Like an Agent
 Dialogue • 15 min
- Learning More & Staying Connected
 Reading • 10 min
- Understanding Agentic AI Concepts
 Graded Assignment • 30 min

Module 2
AI Agents, Tools, Actions, & Language

Module 3
GAME: A Conceptual Framework for AI Agents

Module 4
Agent Tool Mangement

Module 5
Rethinking How Software is Built in the Age of AI Agents

Programmatic Prompting: Building Blocks for AI Agents

Welcome! In this Dialogue, we'll explore programmatic prompting fundamentals – how to send prompts via code, use message roles effectively, and manage memory and structured information – as the foundation for building autonomous AI agents.

Here's what we'll cover:

- **From Manual Chat to Programmatic Prompting:** Understand the shift from manually typing prompts to programmatically sending prompts using code. This automation is the foundation of the agent loop you learned about.
- **The Power of Message Roles:** Explore how system, user, and assistant roles work together. System sets the ground rules and "programs" agent behavior, user provides input, and assistant represents AI responses for conversation context.
- **Memory Management and Structured Information:** Discover how to control what the LLM knows or remembers between iterations and how to send structured information (like JSON) to enable agents to work with APIs and real-world systems.

Need help? Click "I'm stuck" at the top right to get a hint.

When you're ready, click "Start Dialogue"

Start Dialogue

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